

COMP20007 Assignment 2 Task 1C: Written Analysis

Part A: Dynamic Programming Solution

1. State Representation

Each state in the dynamic programming (DP) solution is represented by a tuple $(\text{lake}, \text{time})$, where:

- lake is the current lake the eel is at (an integer from 0 to $N-1$).
- time is the number of days remaining until the breeding season (an integer from 0 to C).

The DP table $\text{dp}[\text{lake}][\text{time}]$ stores the **maximum fat** the eel can have at lake lake with time days remaining.

Auxiliary tables $\text{parent}[\text{lake}][\text{time}]$ and $\text{parentTime}[\text{lake}][\text{time}]$ are used to reconstruct the optimal path.

Fat Management:

- When transitioning between states, fat is updated by subtracting the river traversal cost and adding the fat gain/loss of the destination lake.
- Fat is capped at the maximum allowed (F) and transitions that would result in $\text{fat} \leq 0$ are not allowed.
- Transitions are only allowed if the eel has enough fat to survive the river and does not exceed the time limit.

2. Recurrence

Let $\text{dp}[\text{lake}][\text{time}]$ be the maximum fat achievable at lake lake with time days remaining.

Base Case:

- At the start: $\text{dp}[S][C] = K$ (if $K > 0$), where S is the starting lake, C is the time limit, and K is the initial fat.

Recurrence:

For each lake u , time t , and for each outgoing river from u to v with cost w :

- If $\text{dp}[u][t] \neq -1$ and $t \geq 1$ and $\text{dp}[u][t] > w$:
 - $\text{fatAfterTravel} = \text{dp}[u][t] - w$
 - $\text{fatAfterEating} = \text{fatAfterTravel} + \text{gain}[v]$
 - If $\text{fatAfterEating} > 0$, set $\text{fatAfterEating} = \min(\text{fatAfterEating}, F)$
 - If $\text{dp}[v][t-1] < \text{fatAfterEating}$, then:
 - $\text{dp}[v][t-1] = \text{fatAfterEating}$
 - Update parent pointers for backtrace

Justification:

- The recurrence ensures that at each step, the eel only transitions to valid states (enough fat, within time).
- The DP table always stores the best (maximum) fat for each state.

3. Time Complexity

Let:

- N = number of lakes
- C = time limit
- M = number of rivers (edges)
- D = maximum out-degree (rivers per lake)

Time Complexity:

- For each time step (C), each lake (N), and each outgoing river (D):
 $O(N * C * D)$
- Since $D \leq N$ in the worst case,
 $O(N * C * N) = O(N^2 * C)$

Space Complexity:

- The DP and parent tables are $O(N * C)$ each.

4. Backtrace

Location of Final Answer:

- The optimal solution is found by checking all $dp[0][t]$ for the destination lake 0 and all valid times t ($0 \leq t \leq C$).
- The best fat and earliest arrival time are selected.

Reconstruction:

- Starting from the best $(0, t)$, use the `parent` and `parentTime` tables to trace back through the states until reaching the origin.
- The path is then reversed to produce the correct order.

Part B: Differences from Part A

State Representation

- The state is still $(lake, time)$, but **time now decreases by the river cost** (not always 1).
- When traversing a river with cost w , the time decreases by w instead of 1.

Recurrence

- The recurrence is modified so that for each river with cost w , the transition is only allowed if $t \geq w$, and the next state is at $timeAfter = t - w$.

Time Complexity

- The time complexity remains $O(N^2 * C)$, as the number of possible states and transitions is unchanged, but the time decrement per transition is variable.

Backtrace

- The backtrace process is unchanged, except that time steps may decrease by more than 1 per transition.
-

Summary

- The DP solution efficiently finds the optimal path and fat reserves for the eel, respecting all constraints.
- Part B only changes the time decrement per river, but the overall structure and complexity remain the same.